

守正创新,智引未来:人工智能时代医学人才培养模式的变革与思考

栗 荐, 金 龙

【摘要】 人工智能(AI),特别是生成式模型的迅猛发展,正以前所未有的深度和广度融入临床实践,对医学教育构成了划时代的挑战与机遇。在“健康中国 2030”和国家人工智能战略的背景下,我们必须对医师培养模式进行根本性重塑,将数据驱动的科技素养与以人为本的医学人文传统进行有机融合。该文首先描绘了 AI 在国内外医学教育中的应用现状,特别聚焦于诊断放射学及介入放射学的变革性实践,进而系统性地探讨了医学教育中“应变”与“不变”的核心要素。并提出未来医生的核心角色并非被技术取代,而是成为一名“终极整合者”,即能够娴熟地驾驭 AI 洞见,并将其与临床经验、人文关怀相结合的复合型专家。通过深入分析,该文为构建面向未来的“新医科”课程体系提供了一份务实的路线图,旨在培养出既精通技术又饱含人文精神的新一代临床医师,确保在 AI 时代,医学的本质得以传承与升华。

【关键词】 人工智能;医学教育;介入放射学

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Upholding fundamentals while embracing innovation: reform and reflection on medical talent cultivation in the era of artificial intelligence LI Jian, JIN Long, Department of Interventional Radiology, Affiliated Beijing Friendship Hospital, Capital Medical University, Beijing 100050, China

Corresponding author: JIN Long, E-mail: longjinbfhir@gmail.com

【Abstract】 The meteorically rising of artificial intelligence (AI), particularly the rapid development of generative models, is being integrated into clinical practice with unprecedented depth and breadth, posing an epoch-making challenge and opportunity for medical education. In the context of “Healthy China 2030” and the national AI strategy, the medical educators must fundamentally reshape the physician training mode, aiming to organically integrate data-driven technological literacy with the people-oriented medical humanities tradition. This article introduces the current application status of AI in medical education at home and abroad, with a particular focus on the transformative practice of both diagnostic radiology and interventional radiology. At the same time, it systematically explores the core elements of “embracing innovation bravely” and “upholding fundamentals firmly” in medical education. This article points out that in the future, the core role of physicians is not to be replaced by technology, but to become the “ultimate integrator” of healthcare. This means that doctors need to become composite experts who are not only proficient in mastering AI insights, but also able to combine them with clinical experience and humanistic care. Through in-depth analysis, this article provides a practical roadmap for building a future oriented “new medicine” curriculum, aiming to cultivate a new generation of clinical physicians who are both proficient in technology and full of humanistic spirit, so as to ensure that the essence of medicine can be inherited and elevated in the era of artificial intelligence.

【Key words】 artificial intelligence; medical education; interventional radiology

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作者单位: 100050 北京 首都医科大学附属北京友谊医院放射介入科

通信作者: 金 龙 E-mail: longjinbfhir@gmail.com

1 医学发展的新拐点与中国语境

人工智能(AI)正从放射、病理等辅助诊断领域的“幕后英雄”,迅速走向临床决策、病历书写乃至医患沟通的“台前伙伴”^[1-3]。特别是 2023 年以来,以大语言模型(LLM)为代表的生成式 AI 展现出接近专家的临床推理能力^[4-7],标志着医学发展的又一个历史性拐点。这一变革与我国正在大力推进的“健康中国 2030”规划纲要及《新一代人工智能发展规划》形成了强烈的时代共振^[8-9]。因此对于当代医学教育而言,这不仅是简单的“增加一门 AI 课程”,而是关乎如何重新定义医师角色、重构培养体系的深刻命题。

2 AI 在医学教育中的当前应用

近年来,AI 在医学教育中的应用日益成熟广泛,从多个维度重塑了医学人才培养过程,人工智能与医学教育的融合已呈显著趋势。

放眼全球,领先的医学院校与机构已在教学的核心环节中深度集成 AI 技术^[10-14],以期革新传统的培养模式。在交互式虚拟病人(virtual patients, VP)与诊断训练方面,美国卡耐基梅隆大学、普林斯顿大学等多个团队陆续开发了一系列医患交互型“智能体”(Agent):通过模拟具有复杂病史、个性和情绪的虚拟病人,让医学生能像与真人对话一样,对各类 VP 进行问诊、病史采集、鉴别诊断和沟通技巧等;这类系统还能提供即时、个性化的反馈,指出学生在提问逻辑或共情表达上的不足^[15-17]。与此同时,在 AI 驱动的个性化自适应学习平台(AI-powered adaptive learning platforms, ALP)的构建上,英美、印度等地的基础医学教育正在实验性推广相关应用^[18-19]。这些 ALP 应用能够根据学生在答题中的表现,实时分析其知识薄弱点,并动态推送相关的学习资源(如视频、图表、练习题),实现“千人千面”的个性化复习路径,显著提升学习效率^[20]。另外,在临床技能的智能考核领域,传统上对操作技能(如缝合、穿刺)的评估高度依赖导师的主观印象。而如今,已有项目利用计算机视觉和机器学习手段,对学生的操作录像进行逐帧分析,客观量化各步骤的流畅度、精确度和关键操作的完成情况,实时提供客观标准化的反馈,从而辅助教师进行更精准的专向指导^[21-23]。

在我国,AI 与医学教育的融合同样在国家战略指引下加速推进,并涌现出一系列具有本土特色的理论思考与创新实践^[24]。在“AI+医学”整合课程体系建设方面,部分医学院校积极响应“新医科”号

召,率先构建了系统性的“AI+医学”的课程体系^[25]。这一体系并非简单的课程叠加,而是将智能思维与技能深度融入到基础、临床和预防医学的教学全过程。例如,一些高校开设了如《医学人工智能导论》《医疗大数据处理》等核心课程;以西安交通大学为例,该校创新性地引入 Deepseek 智能体辅助教学,鼓励学生参与跨学科科研项目,旨在培养兼具医学专业底蕴与数据科学素养的复合型人才^[26]。在智能化客观结构化临床考试(objective structured clinical examination, OSCE)领域,为进一步提升考核的标准化与公平性,首都医科大学等高校已着手探索在 OSCE 中引入 ChatGPT 等大语言模型技术^[27]。这些举措旨在将考官从繁琐的重复性评分中解放出来,使其更专注于评估学生的临床思维和人文关怀等高阶能力。

上述教学模式的革新在具体的临床专科中体现得尤为深刻。在放射诊断领域,尽管 AI 辅助教学尚未在放射科大规模普及,但已有大量研究提供了积极的实证支持^[28]。两项针对我国放射科医师、住院医师和医学生的调查显示,尽管许多受访者尚不熟悉 AI 在教学中的具体应用,但绝大多数都表现出了对 AI 整合应用的强烈意愿^[29-30]。受访者尤其希望能利用 AI 驱动的在线平台进行影像阅读和报告撰写训练,以期对未来的临床实践做好充分准备。此外,一个纳入 14 项 AI 辅助放射科教学研究的系统性文献综述指出,92.9%的研究证实 AI 培训能显著改善学员表现,包括提高诊断准确性与解释能力(57.2%)、增强学员信心(57.2%)以及提供模拟实践经验(85.7%)等;更有 35.7%的研究报告显示,经 AI 培训后,学员的敏感性、特异性和准确性中位数分别提升至 72%、80%和 81.3%,显著优于培训前水平^[31]。值得注意的是,AI 已不再仅限于临床科研工具,正逐渐成为嵌入到放射科临床工作流程中的“标准配置”^[32-33]。例如,AI 算法能够自动检出并标记肺结节、脑出血或肺栓塞等急症征象,并智能优化阅片优先级。这种工作流的重塑深刻改变了住院医师的培训模式^[29]:撰写初版报告不再是从零开始的病灶搜寻,而是转变为在 AI 提示下进行复核、鉴别“真伪”并完善诊断逻辑和报告撰写的过程。这要求新时代的医学影像学教育必须着重培养一种新技能:既要善用 AI 提高效率,又要警惕“自动化偏见”(automation bias),具有识别 AI 的漏诊与误报的独立判断能力^[34]。

与侧重影像判读的放射诊断不同,介入放射学除了要求深厚的解剖知识与敏锐的阅片能力之外,

还对精细的临床操作能力提出了极高要求。与外科等操作型专业类似,传统的“学徒制”教学模式(即在资深医师指导下进行临床实践)虽然有效,但存在效率低、风险高且高度依赖带教医师水平等局限^[35]。计算机视觉、机器学习和增强现实(augmented reality, AR)/虚拟现实(virtual reality, VR)等智能技术的融合,为打破这些局限提供了新契机^[36-39]。其中, AI 赋能的高保真模拟训练是介入医学教育中的变革核心。相比缺乏真实触觉反馈的传统物理模拟器^[40-41],结合 AI 算法的血管介入手术机器人与模拟器能够根据导丝与血管壁的接触角度和力度,实时计算并反馈逼真的阻力感^[42]。深度学习模型还能通过分析海量真实手术数据,模拟如严重钙化或迂曲血管的特定触觉特征,显著提升手眼协调训练的真实度^[43]。此外,基于生成对抗网络(generative adversarial network, GAN)的 AI 技术可利用真实的 CT/MRI 数据快速生成无限多样的“虚拟患者 VP”解剖模型^[44-46],加之国内已有利用 AR 技术配合 3D 打印模型降低昂贵耗材教学成本的探索^[47],年轻医师有望在零风险、低成本的环境下,针对罕见解剖变异或高难度病变进行充分练习,积累丰富的“临床”经验。AI 在介入医学教学中的另一大探索是将教学融入实际手术过程,即“术中增强教学”。利用 AI 分割技术,系统可在术中透视图像上实时叠加 3D 解剖结构(如血管走形、肿瘤边界等),帮助低年资医生快速理解二维影像与三维解剖的对应关系^[48]。部分前沿模型甚至能为术者实时预测最佳投照角度或导管路径^[49]。这种“AI 导师”虽然不能替代人类导师,却可作为强大的辅助工具,与带教老师一起完成实时指导和全程安全守护。总而言之,随着 AI 技术深度嵌入从术前规划到术中导航的全流程^[50-52],介入医师的培养目标也随之升级,未来的训练不仅要强调传统的手法技巧,更需着重培养人机协同环境下的决策能力,即如何解读并审慎采信 AI 导航信息,以及在 AI 规划与实际解剖出现偏差时,迅速切换回传统经验并做出正确判断,最终完成精细介入手术的能力^[53-54]。

3 守正:坚守医学教育的人文内核

面对日新月异的技术,医学教育改革愈发需要清醒地认识到技术越是进步,医学的人文属性就越发凸显其不可替代的价值。这些医学的“不变之锚”是保障医疗质量与安全的基石^[55],也是在重塑医学人才培养体系时必须捍卫的底线。医学教育的首要坚守,在于通过言传身教确立医患关系的至高无上

性。共情、倾听、建立信任以及洞察患者言语之外的痛苦与期望,是人类所独有而 AI 无法企及的能力^[56]。当技术可能带来“屏幕”隔阂时,教育应当引导医学生思考,如何通过高质量的“面对面”互动来弥合技术的冰冷,这已成为决定治疗效果与患者满意度的关键。与此同时,扎实的临床基本功构成了对抗算法偏见和技术谬误的最终安全网。详尽的病史采集、规范的体格检查和严谨的临床思维,是医师独立判断的“金标准”。即便 AI 能在数秒内分析完一张胸片,其也无法替代医师通过听诊捕捉到的细微呼吸音变化,或通过触诊感知到的患者的紧张与不适。因此,在智能化教学工具普及的当下,强化这些基本技能的训练反而成为了防止“去技能化”和“自动化偏见”的根本保障。最终,医师的职业精神与伦理担当是永恒的责任。迄今为止,全球范围内任何算法都无法成为法律和伦理上的责任主体,这份责任必须也只能由人类医生承担^[57]。这要求我们的医学伦理教育必须与时俱进,我国的医学 AI 伦理治理研究也必须进一步构建符合国情的框架与监管体系,教育医学生在未来执业中从容应对数据隐私、算法公平性等挑战。

4 创新:重塑 AI 时代的临床核心能力

AI 的深度融入是对医师核心能力的系统性重塑。这迫使医学教育必须完成从传统的“知识灌输者”向“胜任力塑造者”的根本转型。一个核心的转变是医师的角色正从单纯的“知识记忆者”演变为“信息审辨者”。这意味着教学目标中必须新增针对人工智能的精准提问(prompt engineering)^[58]、批判性评估并验证 AI 生成信息真伪的能力。例如,当一个大语言模型建议某个罕见病诊断时,医学生必须具备循证医学的素养,能追溯其证据来源,辨别其是否为“幻觉”(hallucination)^[59]。此外,单打独斗式的诊疗思维正在向“人机协作的指挥者”模式转变,未来的医学教育需要训练学生如何将 AI 视为一个强大的“认知辅助”而非替代品。一位介入医师不能盲目依赖 AI 规划的“最优”穿刺路径,而应将其视为导航参考之一。在进行复杂的肿瘤消融或血管栓塞时,医师需结合患者实时的呼吸配合度、术中导管的触感反馈以及邻近重要脏器的解剖变异,动态修正操作策略,做出最终的执行判断。这种协同也延伸至日常工作流中,例如随着智能语音病历等技术的普及,医师将从繁重的文书工作中解放出来,但其角色也随之演变为“临床数据官”——他们必须学会高效地审核、修订并签署由 AI 生成的医疗文

书,确保每一份记录的准确、完整与合规,并对所有内容负责。这种“人机协同”的新型素养,将成为衡量未来医学教育质量的重要标尺。

5 融合与实践:构建面向未来的“数智介入”教学方案

将上述“守正”与“创新”的理念落到实处,需要一个系统性的顶层设计与实践路径,而这与我国近年来大力倡导的“新医科”建设理念高度契合。“新医科”的核心即是推动医学教育与人工智能、大数据等前沿科技的深度融合。鉴于介入放射学兼具高度依赖影像决策与精细手术操作的双重属性,其是践行这一融合的最佳试验田。因此,我们提出以介入放射学为例,构建分层递进、虚实结合的“新医科”课程与培养体系。

首先,重构课程体系,实现从传统教学向“数智介入”的转型。传统的介入教学往往侧重于解剖记忆与导管操作,而在“新医科”框架下课程设置需更加注重医工交叉。在研究生早期或规培阶段,增设如《介入影像组学与 AI 导论》等理论课程。不同于通识性的传统计算机课,该课程应聚焦于解析 AI 如何进行血管分割、病灶识别以及路径规划。学生需掌握基本的算法评估指标(如 Dice 系数、IOU),学会辨别算法在处理复杂解剖结构(如血管变异)时的局限性。而在进阶教育阶段(如专科医师培训),教学大纲应包含如何利用 AI 软件进行术前 3D 重建、模拟支架释放以及预测栓塞终点。通过案例教学,让学生在面对高难度手术(如 TIPS、复杂动脉瘤)时,习惯于“先在数字孪生模型上预演,后在真实患者身上操作”的新范式^[60]。

其次,建立基于 AI 技术的“全周期”实训路径,以解决介入医学操作风险高、学习曲线陡峭的痛点。利用 AI 赋能的高保真血管介入手术模拟器,建立标准化的“准入机制”。在接触真实病人前,学员必须在模拟器上完成规定学时的训练。系统将自动记录导丝操作的平稳度、透视时间及造影剂用量,并生成多维度的能力分析报告,只有达到特定分数阈值方可进入临床轮转。在实际临床带教中,应推广探索“增强现实(AR)+AI 导师”模式^[61]。利用 AR 眼镜将 AI 重建的 3D 血管树叠加于术野之上,帮助低年资医生建立空间感。同时,带教老师不再是唯一的指导者,AI 系统作为“第二导师”,可实时提示潜在的操作风险,形成“人-机-师”三位一体的教学闭环^[61]。

最后,在新时代背景下,更要强调伦理与规范先

行的底线思维,在教学中必须深度融入严格的 AI 伦理规范,明确界定 AI 建议与医生决策的责任边界;教育学生在享受技术便利的同时,始终保持对生命的敬畏,警惕算法偏见对医疗公平性的潜在影响,特别是针对我国基层医疗资源差异的现实,思考技术下沉的可行性与伦理挑战^[62-63]。

6 结语

AI 时代的医学教育变革,绝非单纯的技术叠加或课程增设,而是一场关乎育人理念、教学模式与评价体系的深刻重塑。展望未来,我们期待通过这种理论与实践相结合、医工交叉的培养模式,造就一批既具备深厚人文情怀与扎实临床功底,又精通数据科学与智能技术的复合型医学人才。只有这样,才能在算法与数据的洪流中,始终把握医学的“温度”,真正实现以智慧引领未来,为“健康中国”战略的深入实施注入源源不断的创新活力。

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